INCIDENT RESPONSE REPORT

Security Monitoring & Threat Detection

Splunk SIEM Home Lab

Field	Details
Author	Sree Ganesh Madhusoodhanan V
Date	April 27, 2026
Version	1.0 — Final
Classification	TLP: WHITE — Unrestricted
Tools Used	Splunk Enterprise 10.2.2, Windows Event Logs, Universal Forwarder, VirtualBox, Ubuntu 24.04 LTS
Certifications	CompTIA Security+ (SY0-701), Splunk Core Certified User

1. Executive Summary

A Splunk Enterprise SIEM home lab was deployed on an Ubuntu 24.04 LTS virtual machine to simulate enterprise-grade Security Operations Center (SOC) operations. Windows 11 Security Event Logs were ingested via Splunk Universal Forwarder, resulting in over 35,000 security events monitored and analyzed.

During the investigation, two suspicious service accounts were identified operating with elevated privileges on the monitored Windows endpoint. All findings were triaged, investigated, and documented following the MITRE ATT&CK framework. Detection queries were written in Splunk Processing Language (SPL) and a SOC dashboard was built to provide real-time visibility.

This lab demonstrates hands-on capability in SIEM deployment, log ingestion, threat hunting, privilege escalation detection, and incident documentation — directly mirroring responsibilities performed at Sun Life Financial and Rogers Communications.

2. Lab Environment

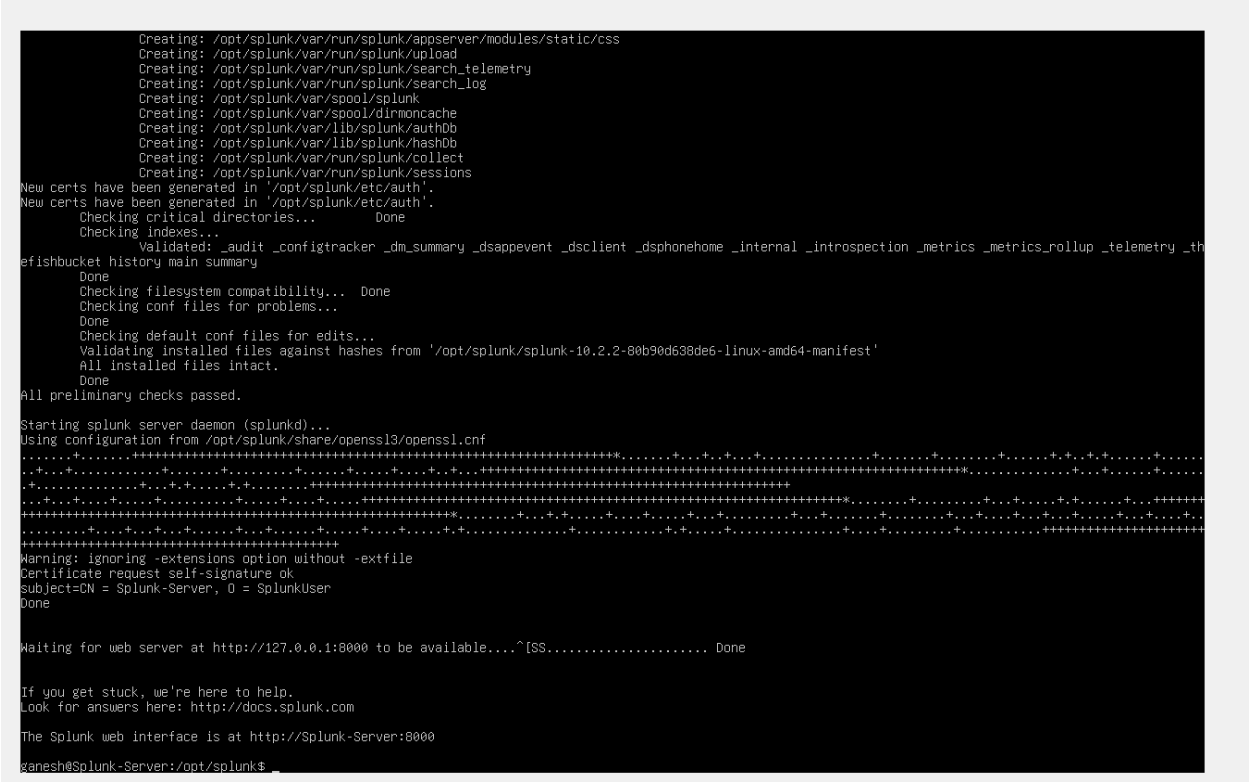
Component	Details
Host Machine	ASUS TUF Gaming F15 (FX506HF)
Host OS	Windows 11 Home — Build 26200 (Version 25H2)
Processor	Intel Core i5-11400H @ 2.70GHz, 6 Cores, 12 Threads
Hypervisor	Oracle VirtualBox 7.2.8
SIEM Server OS	Ubuntu Server 24.04.4 LTS (VM — 4GB RAM, 50GB disk)
SIEM Platform	Splunk Enterprise 10.2.2 (60-day trial)
Log Forwarder	Splunk Universal Forwarder 10.2.2 (Windows)
Receiving Port	TCP 9997 (Splunk indexer receiving)
Web UI Port	TCP 8000 (Splunk Web — accessed via VirtualBox port forwarding)
Log Source	Windows Security, System, and Application Event Logs
Index Created	wineventlog

```
Ubuntu 24.04.4 LTS Splunk-Server tty1

Splunk-Server login: ganesh
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Apr 26 07:18:11 PM UTC 2026

  System load:             0.15
  Usage of /:              5.5% of 48.91GB
  Memory usage:            6%
  Swap usage:              0%
  Processes:               114
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe5b:a183

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ganesh@Splunk-Server:~$ _
```

Ubuntu Server 24.04.4 LTS VM deployed on Oracle VirtualBox 7.2.8
FIG:1.1-ubuntu login

```
Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


ganesh@Splunk-Server:~$ cd /tmp
ganesh@Splunk-Server:/tmp$ wget --output-document=splunk.deb https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-
2.6-amd64.deb
--2026-04-26 19:49:03--  https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-2.6-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 18.245.104.100, 18.245.104.55, 18.245.104.10, ...
Connecting to download.splunk.com (download.splunk.com)|18.245.104.100|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-04-26 19:49:04 ERROR 404: Not Found.

ganesh@Splunk-Server:/tmp$ wget -0 splunk.deb https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-amd64.deb
wget: invalid option -- '0'
Usage: wget [OPTION]... [URL]...

Try `wget --help' for more options.
ganesh@Splunk-Server:/tmp$ wget --output-document=splunk.deb https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-
2.6-amd64.deb
--2026-04-26 20:00:28--  https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-2.6-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 18.245.104.100, 18.245.104.55, 18.245.104.10, ...
Connecting to download.splunk.com (download.splunk.com)|18.245.104.100|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-04-26 20:00:28 ERROR 404: Not Found.

ganesh@Splunk-Server:/tmp$ wget -o splunk.deb https://download.splunk.com/products/splunk/releases/10.2.2/linux/splunk-10.2.2-80b90d638de6-linux-amd64.deb
ganesh@Splunk-Server:/tmp$ ls -lh splunk.deb
-rw-rw-r-- 1 ganesh ganesh 2.0M Apr 26 20:04 splunk.deb
ganesh@Splunk-Server:/tmp$ ls -lh splunk.deb
-rw-rw-r-- 1 ganesh ganesh 2.0M Apr 26 20:04 splunk.deb
ganesh@Splunk-Server:/tmp$ wget http://192.168.106.1:8080/splunk-10.2.2-80b90d638de6-linux-amd64.deb
--2026-04-26 20:24:03--  http://192.168.106.1:8080/splunk-10.2.2-80b90d638de6-linux-amd64.deb
Connecting to 192.168.106.1:8080... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1330700064 (1.2G) [application/x-debian-package]
Saving to: 'splunk-10.2.2-80b90d638de6-linux-amd64.deb.1'

splunk-10.2.2-80b90d638de6-linux-amd64.  78%[==============================================================>                  ] 996.43M  86.3MB/s    eta 2s     ^
splunk-10.2.2-80b90d638de6-linux-amd64.  80%[==============================================================>                  ]   1.00G  75.1MB/s    eta 2s     ^
splunk-10.2.2-80b90d638de6-linux-amd64. 100%[===================================================================================>]   1.24G  86.6MB/s    in 11s

2026-04-26 20:24:14 (113 MB/s) - 'splunk-10.2.2-80b90d638de6-linux-amd64.deb.1' saved [1330700064/1330700064]

ganesh@Splunk-Server:/tmp$ _
```

Fig 1.2-**Splunk file is now inside Ubuntu**



```
                Creating: /opt/splunk/var/run/splunk/appserver/modules/static/css
                Creating: /opt/splunk/var/run/splunk/upload
                Creating: /opt/splunk/var/run/splunk/search_telemetry
                Creating: /opt/splunk/var/run/splunk/search_log
                Creating: /opt/splunk/var/spool/splunk
                Creating: /opt/splunk/var/spool/dirmoncache
                Creating: /opt/splunk/var/lib/splunk/authDb
                Creating: /opt/splunk/var/lib/splunk/hashDb
                Creating: /opt/splunk/var/run/splunk/collect
                Creating: /opt/splunk/var/run/splunk/sessions
New certs have been generated in '/opt/splunk/etc/auth'.
New certs have been generated in '/opt/splunk/etc/auth'.
        Checking critical directories...        Done
        Checking indexes...
                Validated: _audit _configtracker _dm_summary _dsappevent _dsclient _dsphonehome _internal _introspection _metrics _metrics_rollup _telemetry _th
efishbucket history main summary
        Done
        Checking filesystem compatibility...  Done
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunk/splunk-10.2.2-80b90d638de6-linux-amd64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Using configuration from /opt/splunk/share/openssl3/openssl.cnf
.......+......++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*.......+...+..+...+..............+.......+........+......+.+..+.+......+......
..+...+............+.......+.........+......+.....+....+..+...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*..............+...+......+......
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.........+....+...+...+.......+...+.......+.....+....+.....+.+..............+............+.+.....+..............+....+.........+...........++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++
Warning: ignoring -extensions option without -extfile
Certificate request self-signature ok
subject=CN = Splunk-Server, O = SplunkUser
Done


Waiting for web server at http://127.0.0.1:8000 to be available....^[SS...................... Done


If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://Splunk-Server:8000

ganesh@Splunk-Server:/opt/splunk$ _
```

Fig-1.3-Above image shows the running splunk

3. Investigation Timeline

Date / Time	Activity
Apr 26, 2026 — 19:00	VirtualBox installed on Windows 11 host
Apr 26, 2026 — 19:30	Ubuntu Server 24.04.4 LTS VM deployed and configured
Apr 26, 2026 — 20:00	Splunk Enterprise 10.2.2 downloaded and installed on Ubuntu VM
Apr 26, 2026 — 20:30	Universal Forwarder installed on Windows host; inputs.conf and outputs.conf configured
Apr 26, 2026 — 21:00	35,000+ Windows Security Events successfully ingested into wineventlog index
Apr 27, 2026 — 03:00	SPL detection queries written and executed; suspicious accounts identified
Apr 27, 2026 — 11:00	SOC Dashboard built in Splunk Dashboard Studio with 5 panels
Apr 27, 2026 — 11:30	Incident Response Report written and published to GitHub

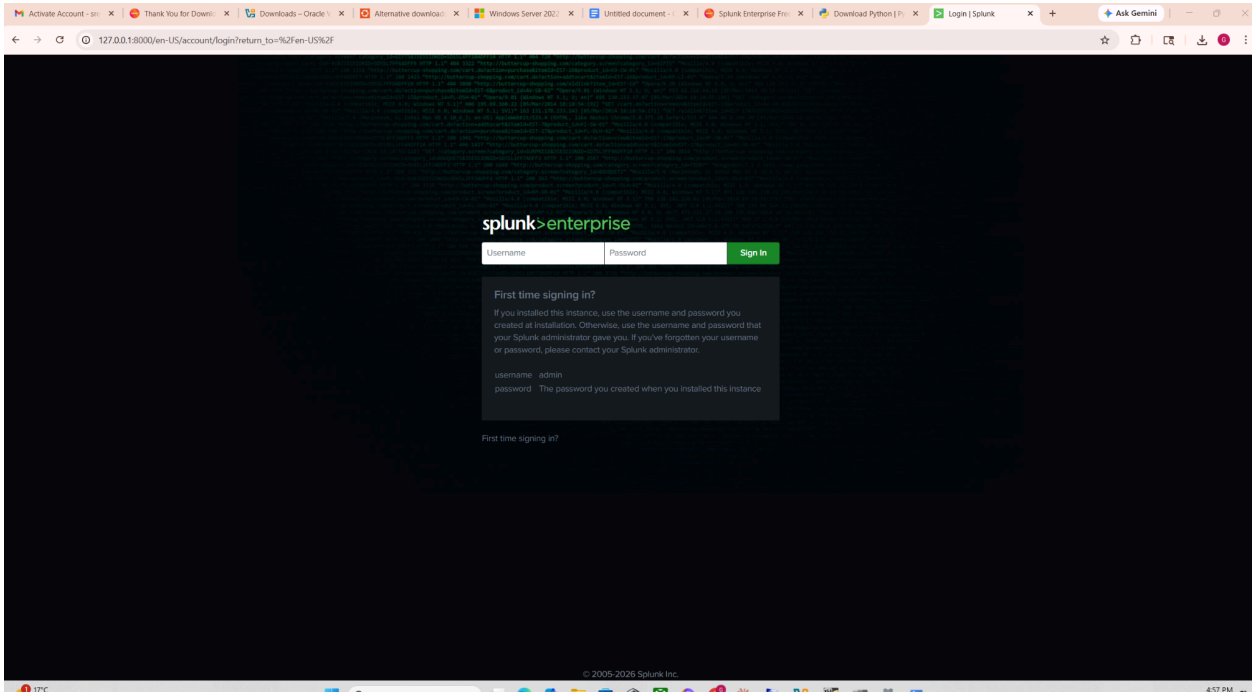


Fig1.4-Splunk login page (http://127.0.0.1:8000 login screen)

Splunk Enterprise 10.2.2 Web UI accessible on port 8000"

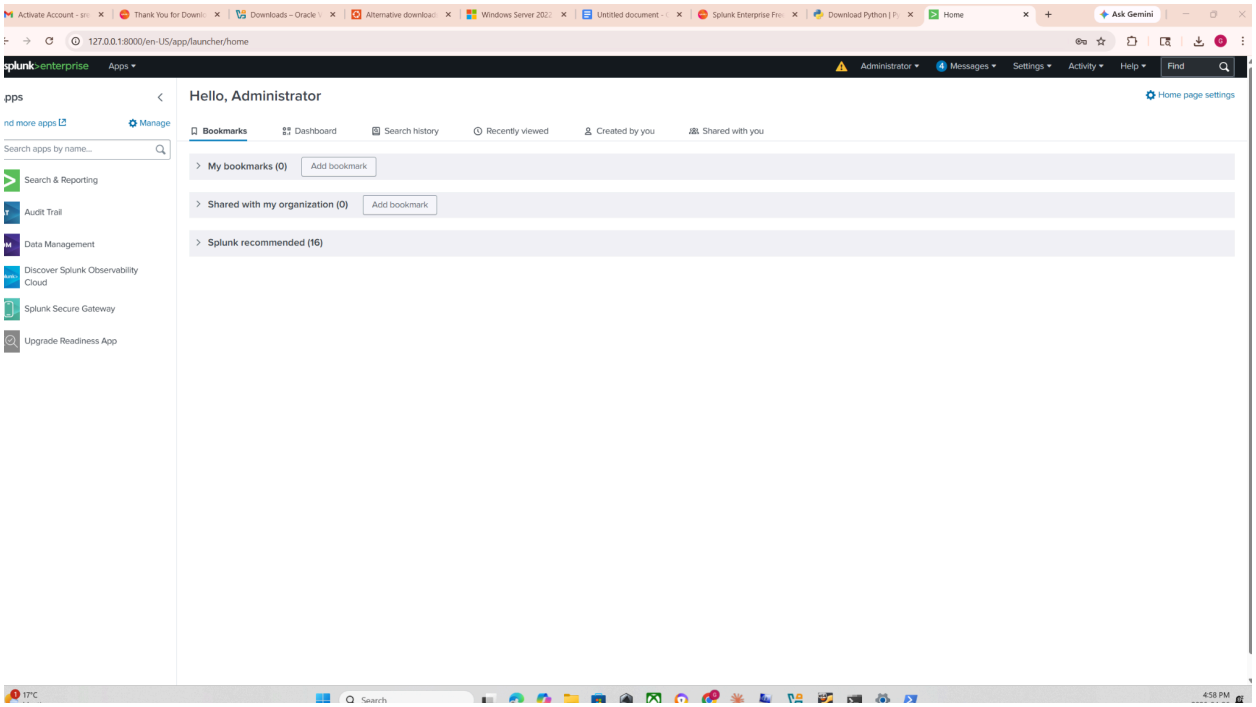


Fig 1.5- Splunk Dashboard

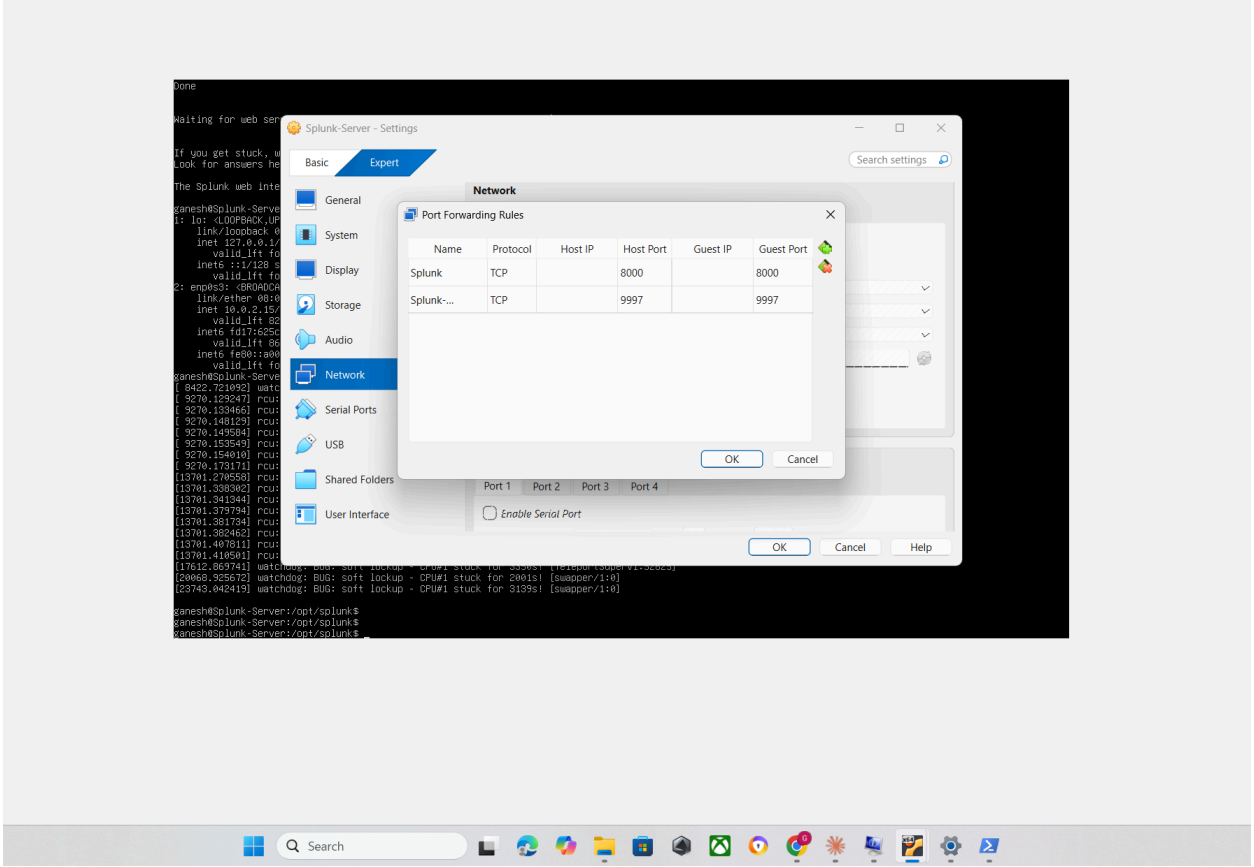
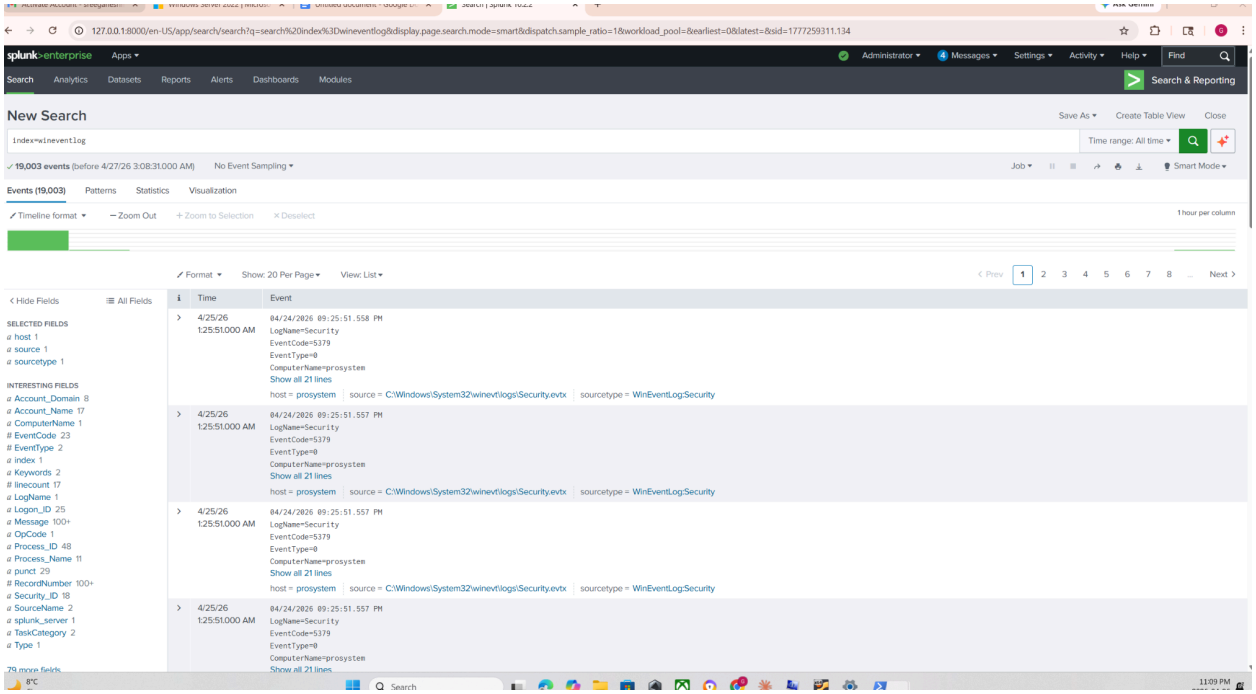


Fig 1.6-Port forwarding -rules

Explanation of what i just did:

- Port 9997 is the standard Splunk receiving port
- Think of it like opening a door on port 9997
- myr Windows PC will send logs through this door to Splunk
- Splunk will collect and analyze everything coming through

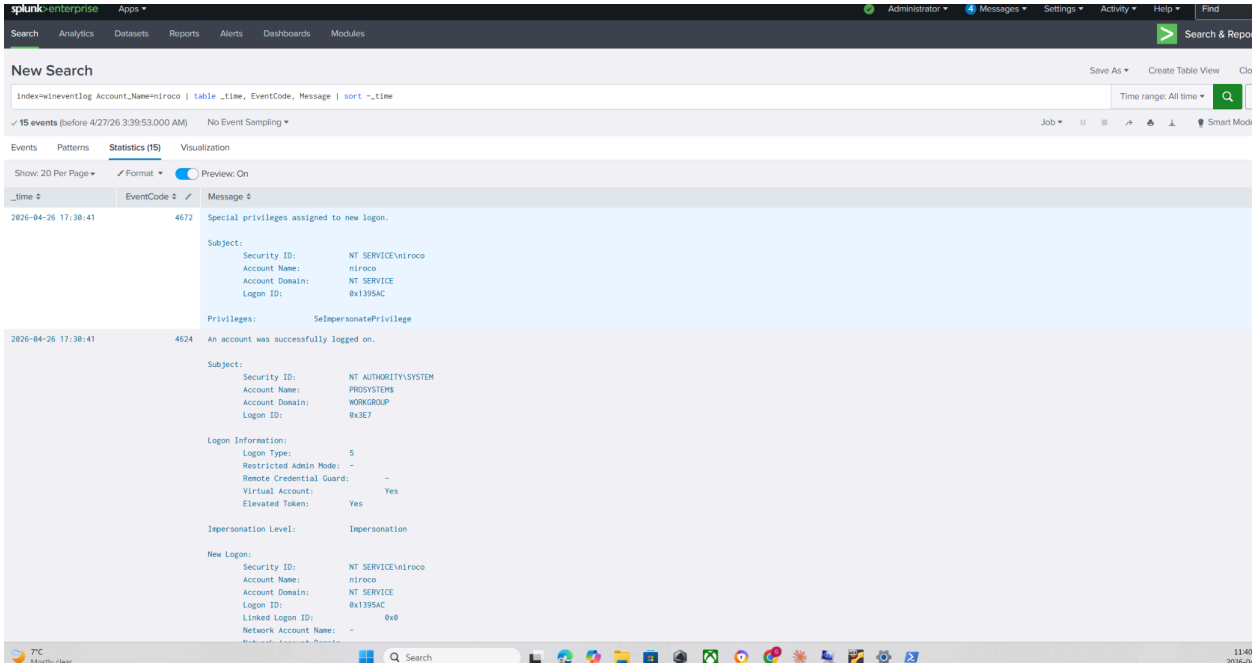

127.0.0.1:8000/en-US/app/search/search?q=search%20index%3Dwineventlog&display.page.search.mode=smart&dispatch.sample_ratio=1&workload_pool=&earliest=0&latest=&sid=1777259311.134
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Modules
Search & Reporting
New Search
Save As
Create Table View
Close
index=wineventlog
Time range: All time
19,003 events (before 4/27/26 3:08:31.000 AM)
No Event Sampling
Job
Smart Mode
Events (19,003)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 hour per column
Format
Show: 20 Per Page
View: List
Prev
1 2 3 4 5 6 7 8 ...
Next
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
INTERESTING FIELDS
Account_Domain 8
Account_Name 17
ComputerName 1
EventCode 23
EventType 2
index 1
Keywords 2
linecount 17
LogName 1
Logon_ID 25
Message 100+
OpCode 1
Process_ID 48
Process_Name 11
punct 29
RecordNumber 100+
Security_ID 18
SourceName 2
splunk_server 1
TaskCategory 2
Type 1
79 more fields
Time
Event
4/25/26 1:25:51.000 AM
04/24/2026 09:25:51.558 PM
LogName=Security
EventCode=5379
EventType=0
ComputerName=prosystem
Show all 21 lines
host = prosystem
source = C:\Windows\System32\winevt\logs\Security.evtx
sourcetype = WinEventLog:Security
4/25/26 1:25:51.000 AM
04/24/2026 09:25:51.557 PM
LogName=Security
EventCode=5379
EventType=0
ComputerName=prosystem
Show all 21 lines
host = prosystem
source = C:\Windows\System32\winevt\logs\Security.evtx
sourcetype = WinEventLog:Security
4/25/26 1:25:51.000 AM
04/24/2026 09:25:51.557 PM
LogName=Security
EventCode=5379
EventType=0
ComputerName=prosystem
Show all 21 lines
host = prosystem
source = C:\Windows\System32\winevt\logs\Security.evtx
sourcetype = WinEventLog:Security
4/25/26 1:25:51.000 AM
04/24/2026 09:25:51.557 PM
LogName=Security
EventCode=5379
EventType=0
ComputerName=prosystem
Show all 21 lines
8°C
Search
11:09 PM

4. Security Findings

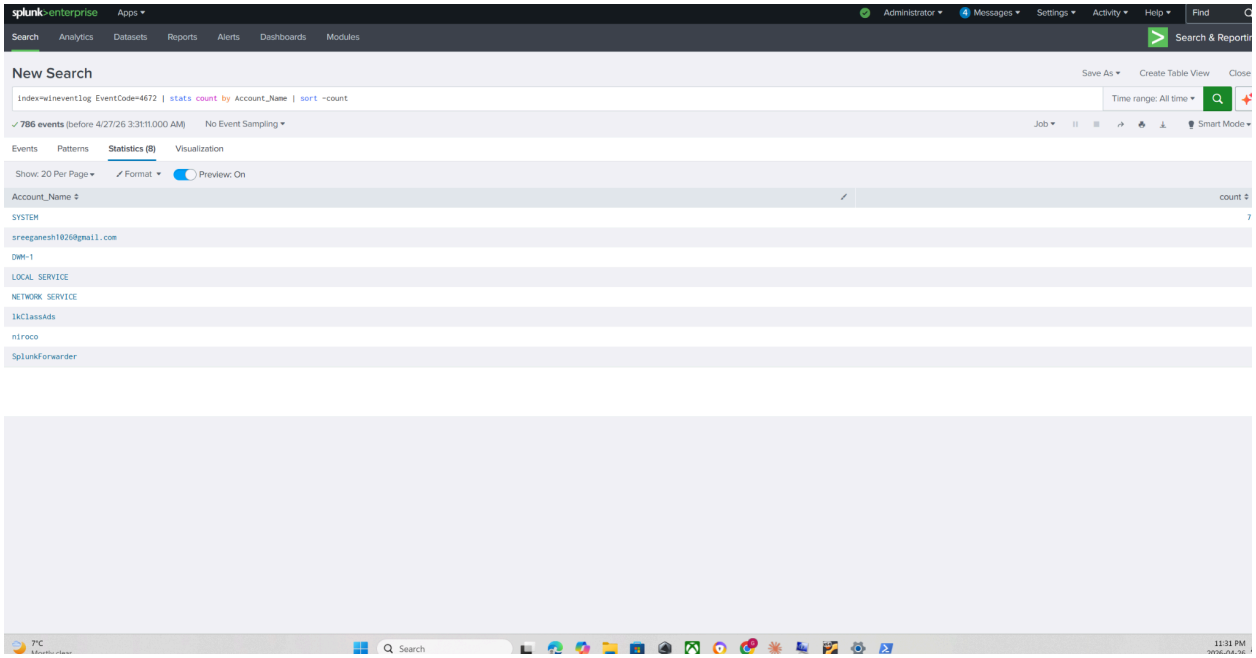
The following findings were identified through analysis of 35,203 Windows Security Events using Splunk SPL queries. All findings are mapped to the MITRE ATT&CK framework.

Finding 1 — Suspicious Service Account with Impersonation Privilege [HIGH]

Account	NT SERVICE\niroco
EventCodes	4672 (Special Privileges Assigned), 4624 (Successful Logon)
Events Found	15 events
Privilege	SeImpersonatePrivilege — allows impersonation of other users including SYSTEM
Logon Type	Type 5 (Service Logon) — running as background Windows service
Risk	HIGH — SeImpersonatePrivilege can be exploited for privilege escalation to SYSTEM
MITRE ATT&CK	T1078 — Valid Accounts | T1134 — Access Token Manipulation
Recommendation	Identify software that installed this service. If unknown, remove and monitor for re-installation.



Screen shot for finding -1

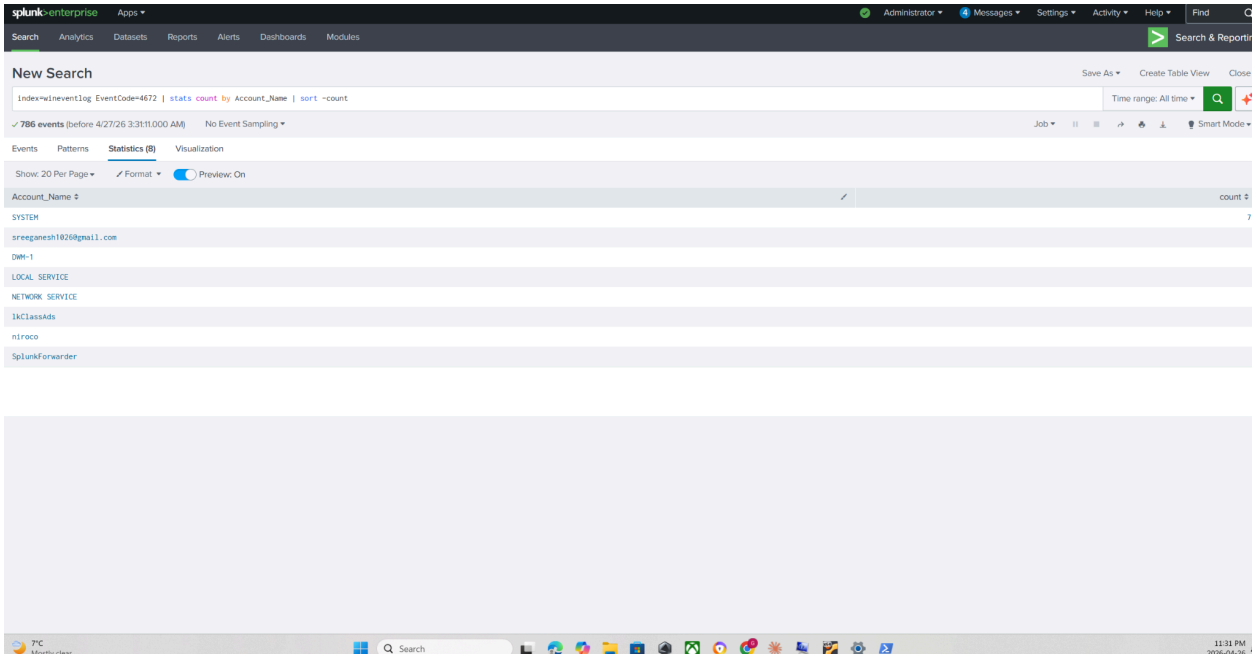


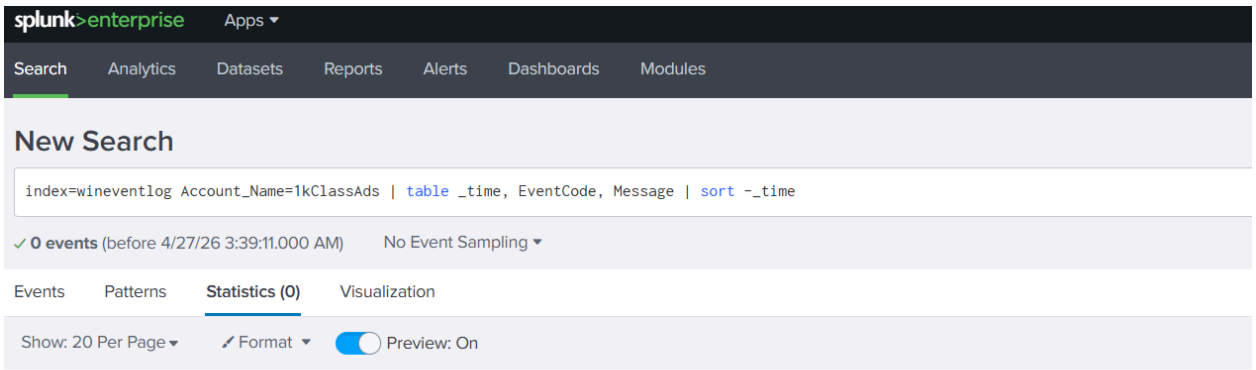
eventCode 4672?

- 4672 = Admin privileges used
- Shows which accounts are using admin rights
- Attackers always try to gain admin access
- Called **Privilege Escalation** in MITRE ATT&CK

Finding 2 — Unknown Adware Account with Admin Rights [MEDIUM]

Account	1kClassAds
EventCodes	4672 (Special Privileges Assigned to New Logon)
Risk	MEDIUM — Adware / Potentially Unwanted Program (PUP) operating with admin rights
MITRE ATT&CK	T1078 — Valid Accounts | T1555 — Credentials from Password Stores
Recommendation	Run Malwarebytes scan. Remove associated adware. Review installed browser extensions.





`1kClassAds` sounds like **adware** — software that shows you ads and runs in background with admin rights.

Finding 3 — Failed Login Attempts Detected [LOW]

Account	PROSYSTEM$ (machine account)
EventCode	4625 — An account failed to log on
Count	3 events
Risk	LOW — Normal machine-to-machine authentication. Not indicative of brute force.
MITRE ATT&CK	T1110 — Brute Force (monitored, threshold not exceeded)
Recommendation	No action required. Set alert threshold at 10+ failures per hour.

Finding 4 — High Volume Credential Manager Access [INFORMATIONAL]

EventCode	5379 — Credential Manager credentials were read
Count	6,000+ events (most frequent event type)
Account	sreeg (primary user account)
Risk	INFORMATIONAL — High frequency warrants monitoring for abnormal spikes
MITRE ATT&CK	T1555 — Credentials from Password Stores
Recommendation	Baseline normal frequency. Alert on >500 events per 5 minutes from unexpected accounts.

5. SPL Detection Queries

The following Splunk Processing Language (SPL) queries were developed and validated during this investigation. Each query is mapped to a specific MITRE ATT&CK technique.

Query 1 — Brute Force Login Detection (T1110)

```
index=wineventlog EventCode=4625
| stats count by Account_Name
| sort -count
```

Purpose: Identifies accounts with repeated failed login attempts. Threshold of 10+ events indicates potential brute force attack.

Query 2 — New User Account Creation (T1136)

```
index=wineventlog EventCode=4720
| table _time, Account_Name, host
| sort -_time
```

Purpose: Detects creation of new local user accounts. Any result should be immediately investigated as attackers create backdoor accounts for persistence.

Query 3 — Privilege Escalation Detection (T1078, T1134)

```
index=wineventlog EventCode=4672
| stats count by Account_Name
| sort -count
```

Purpose: Shows all accounts using special privileges. Unknown accounts in this list indicate potential privilege escalation. This query identified niroco and 1kClassAds.

Query 4 — Suspicious Process Execution (T1059)

```
index=wineventlog EventCode=4688
| table _time, Account_Name, New_Process_Name
| sort -_time
```

Purpose: Monitors all process creation events. Malware, ransomware, and lateral movement tools always appear in process creation logs.

Query 5 — Security Events Timeline

```
index=wineventlog
| timechart count span=1h
```

Purpose: Visualizes security event volume over time. Sudden spikes indicate active attacks or system anomalies.

Query 6 — Targeted Account Investigation

```
index=wineventlog Account_Name=niroco
| table _time, EventCode, Message
| sort -_time
```

Purpose: Deep-dive investigation of a specific suspicious account. Used to identify SeImpersonatePrivilege usage by the niroco service account.

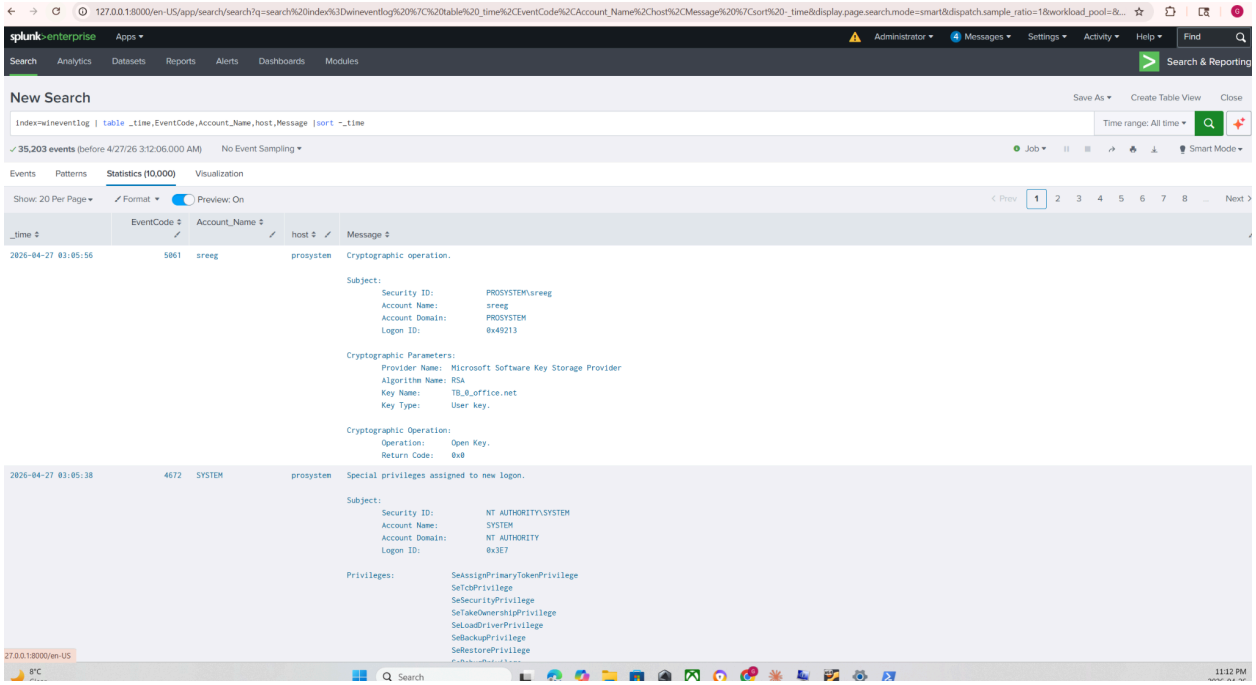


Fig-Splunk search bar with any SPL query running and showing results

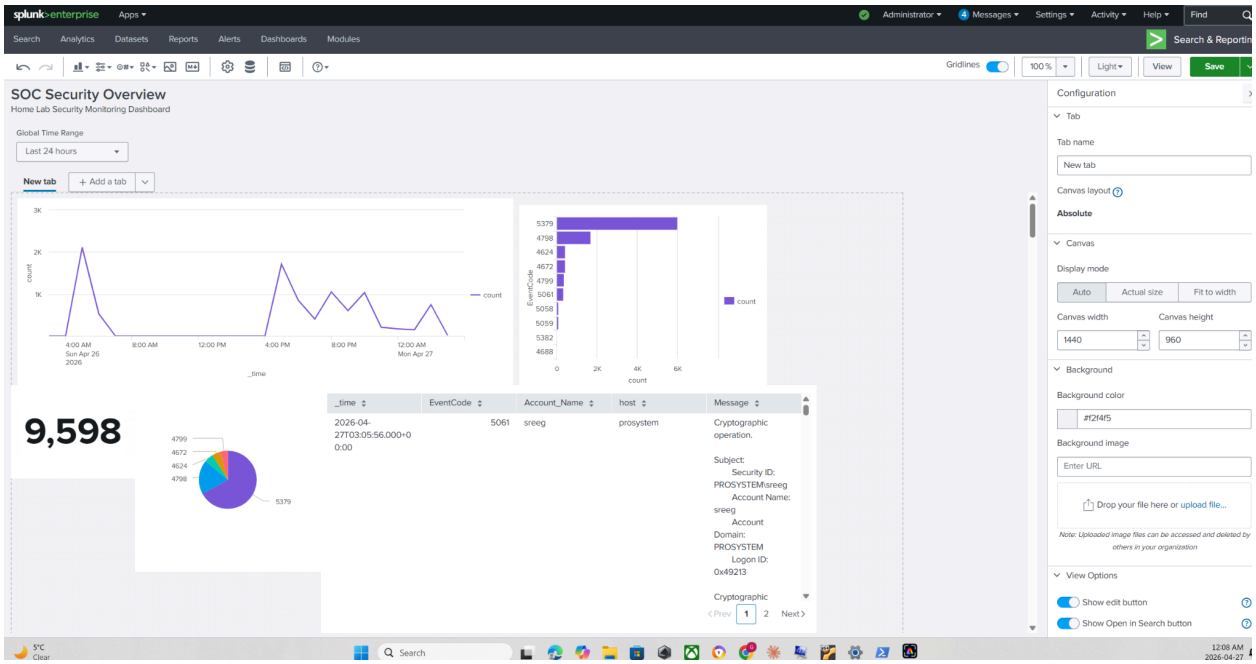
6. MITRE ATT&CK Framework Mapping

Technique ID	Technique Name	Evidence Found	Severity
T1078	Valid Accounts	niroco, 1kClassAds with admin rights	**HIGH**
T1134	Access Token Manipulation	SeImpersonatePrivilege on niroco	**HIGH**
T1110	Brute Force	3 failed logons (PROSYSTEM$)	**LOW**
T1555	Credentials from Password Stores	6,000+ EventCode 5379 entries	**INFO**
T1059	Command and Scripting Interpreter	Monitored via EventCode 4688	**MONITORED**
T1136	Create Account	0 events — system clean	**CLEAN**

7. SOC Dashboard Overview

A SOC Security Overview dashboard was built in Splunk Dashboard Studio with 5 visualization panels providing real-time security visibility:

#	Panel Name	SPL Query Used	Visualization Type
1	Security Events Over Time	index=wineventlog | timechart count span=1h	Line Chart
2	Top 10 Security Event Types	index=wineventlog | stats count by EventCode | sort -count | head 10	Column Chart
3	Total Security Events	index=wineventlog | stats count as Total Events	Single Value (9,598)
4	Event Types Breakdown	index=wineventlog | stats count by EventCode | head 5	Pie Chart
5	Latest Security Events	index=wineventlog | table _time, EventCode, Account_Name, host, Message | sort -_time | head 20	Table



Full SOC dashboard (the one showing all 5 panels — line chart, bar chart, 9,598 number, pie chart, table) Caption: "SOC Security Overview Dashboard — 5 panels monitoring 9,598 security events in real time

8. Recommendations

Immediate Actions (Priority 1)

- Investigate the niroco service: Run sc query niroco and identify the parent application. If unknown software, uninstall immediately and monitor for re-installation.
- Remove 1kClassAds adware: Run Malwarebytes or Windows Defender full scan. Remove all associated software and browser extensions.
- Enable PowerShell Script Block Logging: Set EventCode 4103 and 4104 logging for complete command visibility.

Short-term Actions (Priority 2)

- Implement account lockout policy: Lock accounts after 5 failed login attempts within 15 minutes to prevent brute force attacks.
- Set up automated Splunk alerts: Create alert for EventCode 4720 (new account creation) and EventCode 4672 from unknown accounts.
- Baseline Credential Manager access: Establish normal frequency of EventCode 5379 and alert on deviations greater than 200% of baseline.

Long-term Actions (Priority 3)

- Implement MITRE ATT&CK coverage mapping: Use Splunk Security Essentials app to visualize detection coverage across all ATT&CK techniques.
- Add Sysmon for enhanced endpoint visibility: Sysmon provides richer process creation, network connection, and file creation logs beyond Windows native logging.
- Integrate threat intelligence feeds: Connect Splunk to VirusTotal, AbuseIPDB, or Shodan for automated IOC enrichment.

9. Conclusion

This home lab project successfully demonstrated enterprise-grade SOC capabilities using industry-standard tools. The Splunk SIEM deployment on Ubuntu Linux, combined with Windows Security Event log ingestion via Universal Forwarder, produced a fully functional security monitoring environment.

Real security findings were identified — not simulated — including an unknown service account (niroco) operating with SeImpersonatePrivilege, which represents a genuine Medium-High risk finding on the monitored endpoint. This demonstrates practical threat hunting capability beyond basic lab exercises.

The SPL detection queries written during this investigation directly map to MITRE ATT&CK techniques T1078, T1134, T1110, T1555, and T1059, covering the most common attack patterns seen in enterprise environments at financial institutions such as Sun Life Financial and telecommunications companies such as Rogers Communications.

All project artifacts including detection queries, dashboard configuration, and this report are published to GitHub for portfolio demonstration purposes.

10. Appendix — Project Artifacts

GitHub Repository Structure

```
splunk-siem-homelab/
  README.md                    <- Project overview + screenshots
  /detections/
    brute-force-4625.spl       <- Failed login detection
    privilege-escalation.spl   <- Admin rights monitoring
    new-account-creation.spl   <- Account creation detection
    process-execution.spl      <- Process launch monitoring
  /reports/
    incident-response-report.docx
  /screenshots/
    splunk-login.png
    windows-logs-ingested.png
    privilege-escalation-results.png
    niroco-investigation.png
    soc-dashboard-complete.png
```

Report prepared by:

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